Clustering methods for skipper-CCD images: report

What this report is

Seven clustering methods developed or tested on 2026-10-23..25 for skipper-CCD images, compared with the WADERs / panaSKImg ClusterFinder on: (1) every WADERs-calibrated image on this machine (878 image-CCDs, 16 data sets) with injected objects of known energy and shape; (2) Fe-55 injection at four cluster widths, close pairs and the real K-alpha peak; (3) injected muons, worms, alphas, hot columns, X-ray pairs in real Am-241 images; (4) few-electron events on simulated blanks (efficiency and fakes); (5) the crowded Am-241 CCD (29 images) through the constrained Np L-line fit. Tables are recomputed from the saved results by make_clustering_report.py.

Result in one paragraph

A_GMX (adaptive switch + Gaussian-mixture EM deblending) is the best general clusterer: unbiased Fe-55 energies at every cluster width (WADERs: -8 % at 1.8 px, -26 % on the worst CCD), 2.7x better energy resolution than WADERs, close X-ray pairs resolved (87-93 % vs 0-11 %), tracks kept whole, few-electron events found without fakes, and on the crowded Am-241 CCD Np L-beta 19.9 sigma vs 15.5 with the right energy scale (k = 1.001 vs 0.984). It supersedes MHC, the hybrids and the MHC-engine adaptive. MFD+G stays the fastest choice for isolated X-rays (Fe-55 calibration); WADERs / B auto stay the pipeline reference and the fallback for images whose background is not dark current.

Recommendation by situation

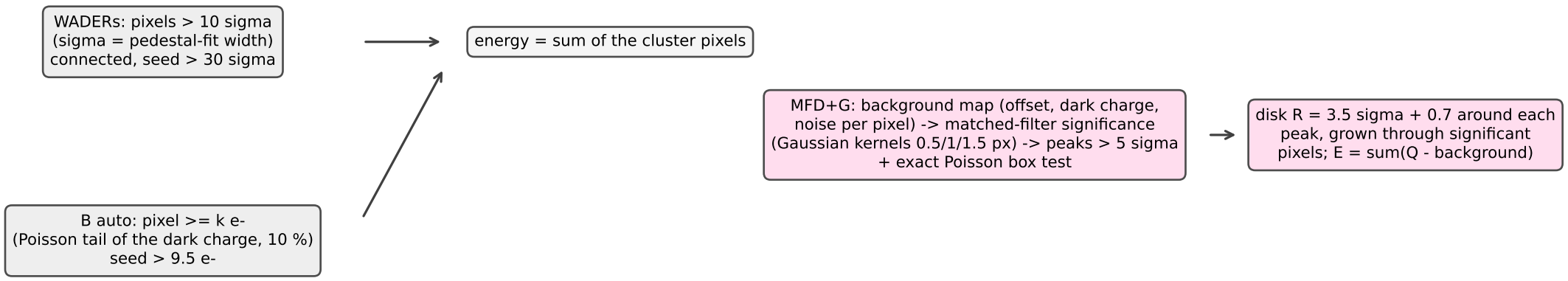
general analysis of any image: A_GMX (adaptive_cluster(..., engine='gmx')), cut blend < 0.02 when purity matters

Fe-55 gain / calibration only: MFD+G (same energies as A_GMX, 5x faster, Fe-55 success 0.972 vs 0.951)

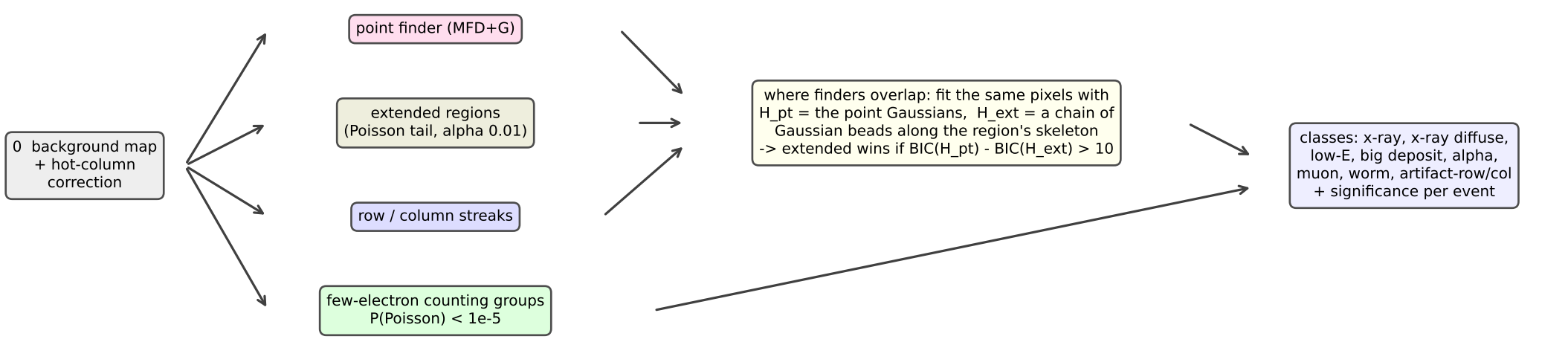
standard pipeline, speed: WADERs or B auto (--cluster-threshold-e auto)

few-electron searches: A_GMX's counting finder (0 fakes; 3 e- at 0.001 e-/pix), or >= 1 e- clustering with its Poisson significance as the fake filter

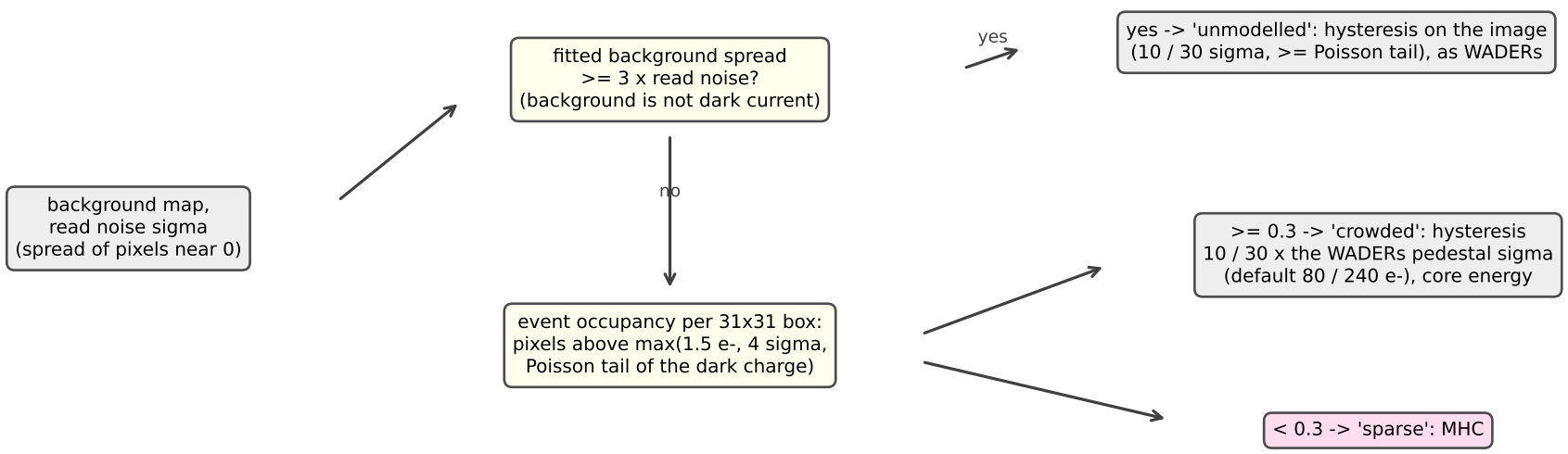
WADERs / B auto (hysteresis) and MFD+G (matched filter)



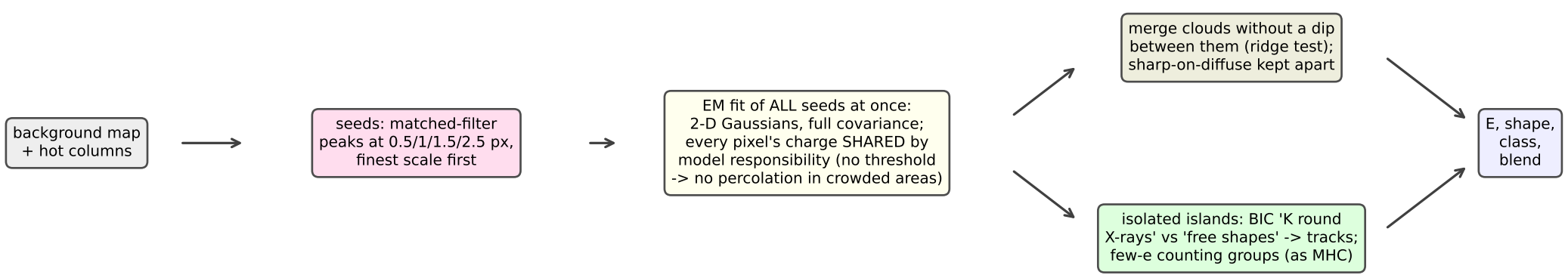
MHC - multi-hypothesis clustering (new)



adaptive - picks the method per region (new)



GMX - Gaussian-mixture EM deblending (from the exploratory agent, verified)



The methods: what they do, where the ideas come from, why they work

WADERs (panaSKImg ClusterFinder) - the reference

What: pixels above n_pix x sigma that touch (8 neighbours) form a cluster; it is kept if one pixel is above n_seed x sigma (10 and 30). sigma is the width of the pedestal fit of the CCD. Energy = sum of the cluster pixels.

Origin: two-threshold (hysteresis) segmentation, as in Canny's edge detector (IEEE PAMI 8, 679, 1986) and the detection / analysis thresholds of SExtractor (Bertin & Arnouts 1996, A&AS 117, 393).

Why it works / fails: simple and fast. Its thresholds are in units of the pedestal sigma, which on noisy or high-dark-current CCDs is 5-15x the read noise (7-21 e-): the tails of diffuse clusters are cut (energies 8 % low at sigma_xy 1.8 px), close X-rays merge, few-electron events are invisible. On a crowded CCD the same inflated sigma (8 e- on Am-241 EXT4) happens to give the right thresholds (80 / 240 e-).

B auto (--cluster-threshold-e auto, in run_WADERs since commit acbac7d)

What: as WADERs but the pixel threshold is set in electrons from the dark charge: smallest k >= 2 with P(Poisson(lambda) >= k) <= 0.10, pixel > k - 0.5 e-, seed > 9.5 e-.

Origin: site percolation (8-neighbour threshold p_c = 0.407): if the fraction of dark-charge pixels above threshold stays far below p_c, noise cannot build connected clusters; a per-pixel false-alarm rate as in FDR source finding (Hopkins et al. 2002, AJ 123, 1086).

Why it works: thresholds follow the physics of the background (Poisson dark charge), not the pedestal fit; recovers the charge of diffuse clusters. Still merges close X-rays and cannot see < 10 e- events (seed 9.5 e-).

MFD+G - matched-filter detection + disk + growth (exploratory agent, 2026-10-23)

What: (1) background map: per-pixel offset, Poisson dark charge and read noise, fitted row by row with clusters masked; (2) matched-filter significance T = sum w (Q - mean) / sqrt(sum w^2 var) with Gaussian kernels of 0.5, 1 and 1.5 px; (3) seeds = local maxima with T > 5 that also pass an exact Poisson test of the 3x3 / 5x5 box sum; (4) centre and width from background-subtracted moments in the seed's Voronoi cell; (5) footprint = disk of radius 3.5 sigma + 0.7 px, grown through pixels above a Poisson-tail threshold (alpha 0.01, morphological reconstruction); (6) energy = sum of (Q - background) over the footprint.

Origin: radio-astronomy source finders (SoFiA: Serra et al. 2015, Westmeier et al. 2021; PowellSnakes: Carvalho et al. 2009; wavdetect: Freeman et al. 2002), the Li & Ma (1983, ApJ 272, 317) counting significance, morphological reconstruction (Vincent 1993, IEEE TIP 2, 176).

Why it works: detection does not depend on a pixel threshold, and the footprint is sized from the cluster's own width, so diffuse clusters keep their tails and the background is subtracted pixel by pixel: unbiased energies at every width and close X-rays resolved. Fails on extended deposits (a track becomes several 'X-rays') and where the background is not dark current (energies come out high).

hybrid v1 / v2 - MFD+G followed by a merge step (2026-10-24)

What: MFD+G, then pieces inside a B-mask region at least 14 px across are merged (v1: all; v2: if a lenient saddle test passes); elsewhere two pieces merge only if the charge on the best path between them never drops below half of the smaller peak (saddle test).

Origin: SExtractor's deblending contrast (Bertin & Arnouts 1996) and the 'dynamics' of watershed / component trees (Grimaud 1992; Najman & Couprie 2006), used in reverse: a track has no deep valley between its pieces.

Why it partly works: joins smooth tracks, keeps compact pairs apart. Superseded by MHC: it merges diffuse X-ray pairs (whose union is > 14 px) and fails on crowded CCDs (everything is one large region).

MHC - multi-hypothesis clustering (new, 2026-10-24, clustering_tests/mhc.py)

What: (0) background map + hot-column correction (column excess of the quiet pixels, subtracted where > 5 sigma); (1) specialised finders, each for one kind of deposit: points (MFD+G), extended regions (connected pixels above a Poisson-tail threshold), row / column streaks (>= 85 % of the charge in <= 2 rows or columns and a run of >= 8 px), few-electron counting groups (pixels >= 1 e-, kept if P(Poisson(local dark charge) >= charge) < 1e-5); (2) arbitration where finders overlap: the same pixels are fitted with H_pt = the point Gaussians and H_ext = a chain of Gaussian beads along the region's skeleton (non-negative amplitudes, Poisson-weighted chi^2, sparse solver); the extended explanation wins if BIC(H_pt) - BIC(H_ext) > 10; (3) a class for every event (x-ray, x-ray diffuse, low-E, big deposit, alpha, muon, worm, artifact-row / -column) and a significance.

Origin: model selection with the Bayesian information criterion (Schwarz 1978, Ann. Stat. 6, 461) as used for source / model choice in astronomy (The Tractor: Lang, Hogg & Mykytyn 2016); skeletonisation (Zhang & Suen 1984); few-electron clustering with Poisson significance as in skipper-CCD dark-matter searches (SENSEI, DAMIC-M).

Why it works: instead of one rule for everything, each deposit type is found by the tool that suits it, and where two explanations compete the data decide with a penalty for extra parameters (a free chain could fit anything; BIC stops it from swallowing close X-ray pairs). Fails where the background is not dark current (inherits MFD+G's energies) and is the slowest method (up to ~80 s on a crowded 1.75 Mpix CCD).

adaptive - picks the method per region (new, 2026-10-25, clustering_tests/adaptive.py)

What: (1) background map and read noise sigma (spread of the pixels near 0 e-); (2) if the fitted background spread is >= 3x the read noise (and >= 0.5 e-), the background is not dark current + noise -> 'unmodelled': hysteresis on the image itself (10 / 30 sigma, never below the Poisson tail), as WADERs; (3) otherwise the event occupancy is measured in 31x31 px boxes (fraction of pixels above max(1.5 e-, 4 sigma, Poisson tail of the dark charge)): boxes with >= 30 % are 'crowded' -> hysteresis at 10 / 30 x the WADERs pedestal sigma (default 80 / 240 e-), the rest 'sparse' -> MHC (engine 'mhc'); with engine 'gmx' GMX clusters the whole image instead.

Origin: regime switching from what was measured in this study (no single rule wins everywhere); locally adaptive thresholds (Niblack 1986; Sauvola & Pietikainen 2000) were tried for the crowded regime and rejected (they fragment the charge sheet).

Why it works: the two failure modes of the model-based methods are recognised from the data (a background that the dark-current model cannot describe; crowding by events, not by dark current) and handed to the method that is best there. The occupancy counts only pixels above the dark-charge tail, so high-dark-current Fe-55 CCDs stay with the model-based method (which is much better than WADERs there).

GMX - Gaussian-mixture EM deblending (exploratory agent, 2026-10-25, clustering_tests/gmx.py)

What: (0) background map + hot columns; (1) seeds: local maxima of the Poisson-calibrated matched filter at 0.5 / 1 / 1.5 / 2.5 px, finest scale first (a coarse maximum is kept only if no finer seed is close); (2) one 2-D Gaussian with a full covariance per seed, all fitted at once by expectation-maximisation with the pixel charge as weight; residual peaks become new Gaussians; insignificant ones are dropped; (3) every pixel's charge is SHARED among the Gaussians in proportion to their model values; (4) Gaussians are merged when the two-cloud model has no dip between them (unless a sharp one sits on a diffuse one); in isolated islands a BIC test 'K round X-rays' vs 'free shapes' joins tracks; (5) energy = shared charge (+ significant uncovered pixels next to a cluster), few-electron groups as MHC, classes as MHC, and 'blend' = charge of other clusters under this one / its own.

Origin: crowded-field photometry in astronomy (DAOPHOT: Stetson 1987, PASP 99, 191; The Tractor; Scarlet: Melchior et al. 2018, A&C 24, 129); EM for mixtures (Dempster, Laird & Rubin 1977, JRSS B 39, 1); merging mixture components by modality / ridgeline (Ray & Lindsay 2005, Ann. Stat. 33, 2042; Hennig 2010, ADAC 4, 3); scale-space detection (Lindeberg 1994).

Why it works: no threshold decides which pixel belongs to which cluster, so a dense field cannot percolate into one blob and neighbours do not steal each other's charge; 'blend' gives a per-cluster purity that can be cut on. This is what beats WADERs on the crowded CCD. Fails where the background

is not dark current (like MFD+G).

A_GMX = adaptive + GMX (recommended, 2026-10-25)

Full description with every step, equations and figures: docs/aGMX/aGMX_clustering.tex (Overleaf: docs/aGMX/aGMX_clustering_overleaf.zip).

What: the adaptive switch (fitted background spread >= 3 x read noise -> WADERs-like hysteresis on the image) in front of GMX on the whole image. Every cluster also gets the event occupancy of its neighbourhood (information).

Why it works: GMX is the best model-based method in every regime where the background model holds (sparse, high-dark-current, crowded), and the switch catches the one regime where it does not.

Rejected ideas (tested)

Track-first then MFD+G (merged close pairs, poor on dense CCDs); saddle merge alone (splits lumpy electron tracks); local-median thresholds and extra energy rings in crowded regions (fragment the sheet / take the neighbours' charge); occupancy mask (removes almost all of a crowded CCD); persistence watershed (PWS, cannot split close pairs); round-Gaussian stamp fits (real front-side X-rays are not round, ~50 % charge lost); mean-shift / DBSCAN (equivalent to thresholds + connectivity).

Wide test: every method on every image

Every WADERs-calibrated image found under ~/Documents (copies removed): 878 image-CCDs. Input = the calibrated image in electrons (linear gain). Per image-CCD: 12 injected Fe-55 clusters (sigma 0.3-1.6 px), 4 muons, 4 worms, 4 pairs of 5.9 keV X-rays 4 px apart, the same for every method. 'ok' = one cluster with E within 5 % (pairs: both resolved within 10 %); rms = robust spread of E_rec / E_true of the Fe-55 clusters.

All image-CCDs

method	Fe-55 ok	E/E_true	rms [%]	muon	worm	pair 4px	s/img
WADERs	0.856	1.0002	0.52	0.59	0.74	0.11	0.0
B auto	0.905	0.9997	0.59	0.62	0.78	0.13	0.0
MFD+G	0.933	1.0002	0.21	0.18	0.37	0.94	0.1
hybrid v1	0.873	1.0001	0.20	0.58	0.66	0.84	0.0
hybrid v2	0.930	1.0001	0.21	0.63	0.68	0.85	0.0
MHC	0.933	1.0001	0.21	0.64	0.77	0.86	1.6
adaptive	0.931	1.0000	0.21	0.64	0.77	0.83	1.3
GMX	0.913	1.0000	0.19	0.63	0.78	0.89	0.6
adaptive+GMX	0.925	0.9999	0.19	0.64	0.79	0.87	0.8

Fe-55 data sets

method	Fe-55 ok	E/E_true	rms [%]	muon	worm	pair 4px	s/img
WADERs	0.880	1.0003	0.47	0.62	0.78	0.11	0.0
B auto	0.927	0.9998	0.53	0.65	0.81	0.13	0.0
MFD+G	0.972	1.0001	0.17	0.20	0.39	0.95	0.1
hybrid v1	0.926	1.0001	0.16	0.63	0.71	0.87	0.0
hybrid v2	0.972	1.0001	0.17	0.68	0.73	0.88	0.0
MHC	0.964	1.0000	0.17	0.68	0.81	0.87	0.9
adaptive	0.961	1.0000	0.17	0.67	0.81	0.87	0.8
GMX	0.951	1.0000	0.16	0.67	0.83	0.91	0.6
adaptive+GMX	0.951	1.0000	0.16	0.67	0.83	0.91	0.6

Am-241 data sets

method	Fe-55 ok	E/E_true	rms [%]	muon	worm	pair 4px	s/img
WADERs	0.830	0.9997	0.77	0.42	0.60	0.12	0.1
B auto	0.822	0.9996	0.89	0.42	0.60	0.14	0.2
MFD+G	0.738	1.0041	1.21	0.11	0.27	0.88	3.8
hybrid v1	0.605	1.0017	0.78	0.29	0.37	0.69	8.1
hybrid v2	0.728	1.0039	1.18	0.37	0.46	0.72	16.9
MHC	0.793	1.0012	1.11	0.41	0.53	0.83	76.7
adaptive	0.794	0.9993	0.95	0.41	0.56	0.61	38.3
GMX	0.730	1.0001	0.75	0.38	0.50	0.74	39.4
adaptive+GMX	0.819	0.9995	0.72	0.42	0.58	0.62	22.7

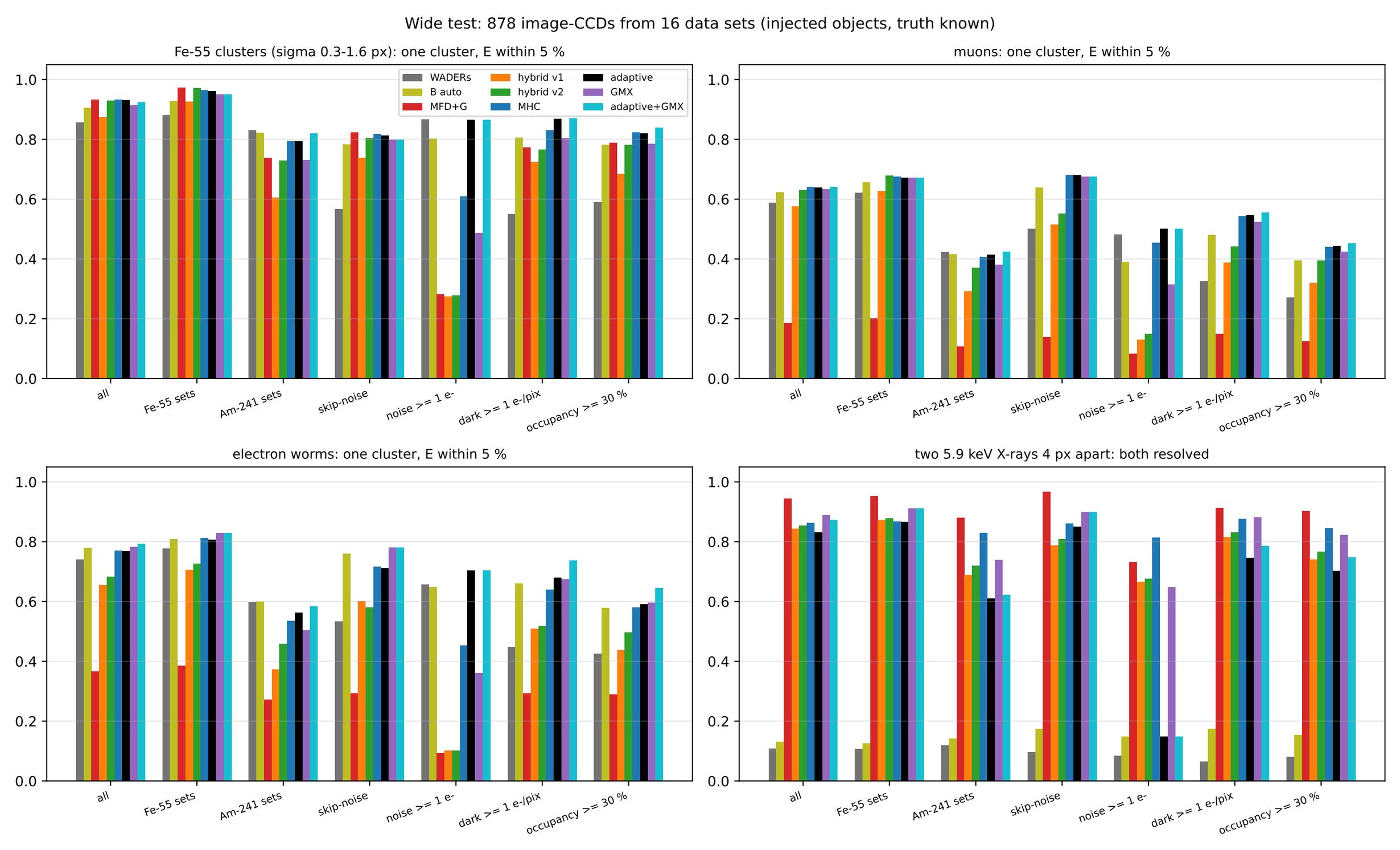
Noisy images (fitted background spread >= 1 e-: PD06/DM12 Am-241)

method	Fe-55 ok	E/E_true	rms [%]	muon	worm	pair 4px	s/img
WADERs	0.867	0.9991	1.66	0.48	0.66	0.08	2.4
B auto	0.802	0.9885	2.08	0.39	0.65	0.15	2.5
MFD+G	0.281	1.0882	9.15	0.08	0.09	0.73	69.8
hybrid v1	0.275	1.0819	8.24	0.13	0.10	0.67	98.4
hybrid v2	0.278	1.0877	8.99	0.15	0.10	0.68	156.0
MHC	0.608	1.0243	5.62	0.45	0.45	0.81	396.6
adaptive	0.864	0.9884	2.02	0.50	0.70	0.15	1.3
GMX	0.488	1.0072	4.45	0.31	0.36	0.65	226.1
adaptive+GMX	0.864	0.9884	2.02	0.50	0.70	0.15	1.3

High dark charge (>= 1 e-/pix)

method	Fe-55 ok	E/E_true	rms [%]	muon	worm	pair 4px	s/img
WADERs	0.550	1.0082	2.40	0.33	0.45	0.07	0.1

B auto	0.806	1.0014	2.10	0.48	0.66	0.17	0.1
MFD+G	0.772	1.0017	1.07	0.15	0.29	0.91	0.7
hybrid v1	0.724	1.0015	1.02	0.39	0.51	0.82	0.1
hybrid v2	0.765	1.0016	1.06	0.44	0.52	0.83	0.3
MHC	0.829	1.0002	1.00	0.54	0.64	0.88	16.2
adaptive	0.868	0.9990	0.94	0.55	0.68	0.75	7.3
GMX	0.804	0.9994	0.88	0.52	0.67	0.88	1.9
adaptive+GMX	0.871	0.9986	0.90	0.55	0.74	0.79	1.6


Wide test: 878 image-CCDs from 16 data sets (injected objects, truth known)
Fe-55 clusters (sigma 0.3-1.6 px): one cluster, E within 5 %
muons: one cluster, E within 5 %
electron worms: one cluster, E within 5 %
two 5.9 keV X-rays 4 px apart: both resolved
WADERs
B auto
MFD+G
hybrid v1
hybrid v2
MHC
adaptive
GMX
adaptive+GMX
0.0
0.2
0.4
0.6
0.8
1.0
all
Fe-55 sets
Am-241 sets
skip-noise
noise >= 1 e-
dark >= 1 e-/pix
occupancy >= 30 %

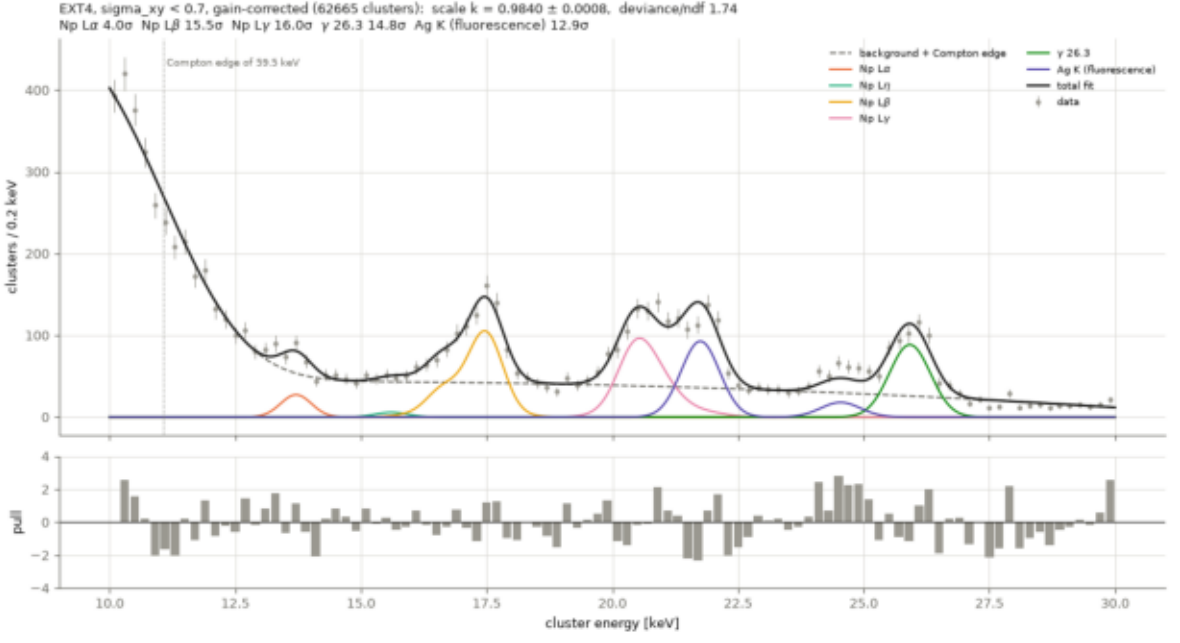
Crowded CCD: Am-241 10-30 keV constrained fit (29 images, EXT4)

Am-241 runs_29img, EXT4 (in one corner every pixel holds > 20 e-), 29 images, gain correction per gain period. Constrained fit of 10-30 keV: Np L lines with fixed intensity ratios (Lepy et al. 2008), one energy scale k, Compton edge, Ag K fluorescence. Z_np = combined significance of La, Lb, Lg and the 26.3 keV gamma.

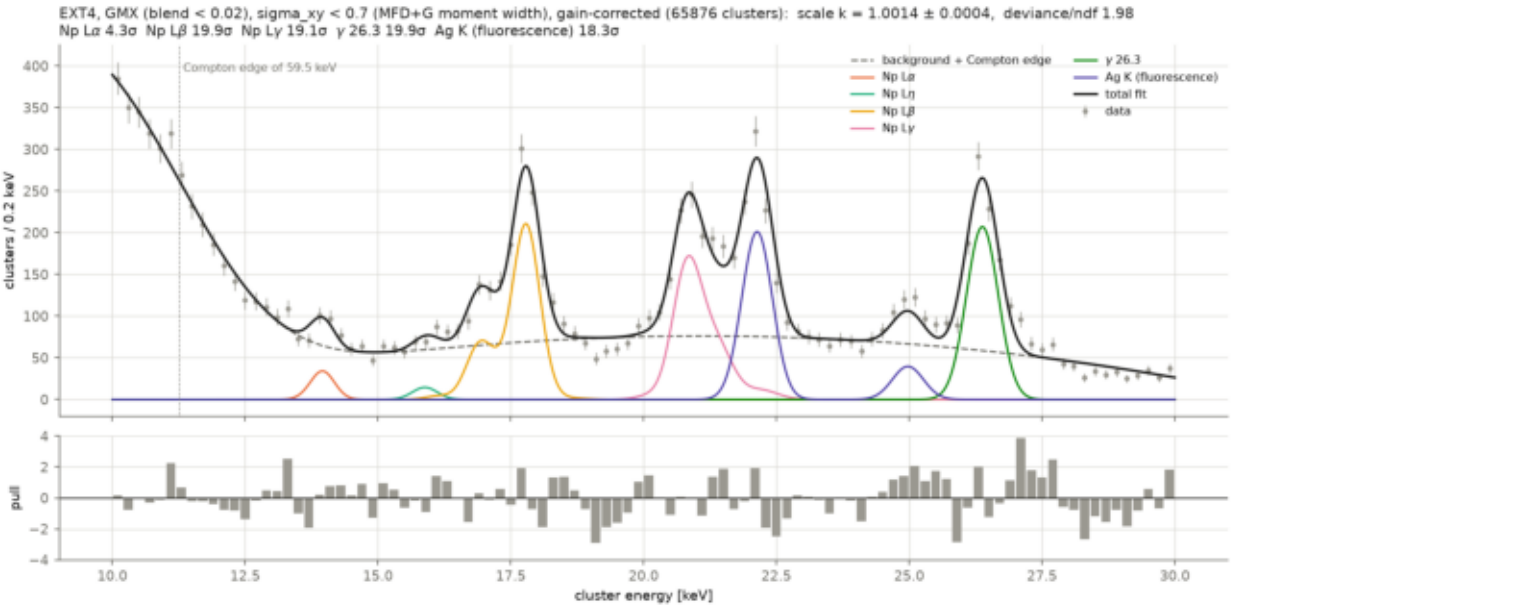
clustering (cut)	clusters	k	dev/ndf	Z_np	La	Lb	Lg	g26	Ag K
- WADERs (sigma_xy < 0.7)	62665	0.9840		1.74	27.1	4.0	15.5	16.0	14.8
12.9 MFD+G (sigma_xy < 0.7)	28811	1.0032		1.14	14.8	2.1	8.2	9.1	8.0
8.5 hybrid v1 (sigma_xy < 0.7)	5270	0.9990		1.31	8.8	0.7	4.4	5.2	5.6
3.5 hybrid v2 (sigma_xy < 0.7)	23862	1.0020		1.14	14.5	1.9	7.9	8.5	8.4
8.4 MHC (sigma_xy < 0.7)	27250	1.0032		1.10	14.8	1.9	8.0	9.1	8.2
8.5 adaptive, MHC engine (sigma_xy < 0.7)	60158	0.9860		1.51	25.5	3.0	14.4	15.4	14.0
12.0 GMX (sigma_xy < 0.7)	120280	1.0007		1.74	25.9	1.3	14.6	15.6	14.6
14.1 GMX (sigma_xy < 0.7, blend < 0.02)	65876	1.0014		1.98	34.3	4.3	19.9	19.1	19.9
18.3 A_GMX (sigma_xy < 0.7)	120280	1.0007		1.74	25.9	1.3	14.6	15.6	14.6
14.1 A_GMX (sigma_xy < 0.7, blend < 0.02)	65876	1.0014		1.98	34.3	4.3	19.9	19.1	19.9
18.3									

Checks: the GMX / A_GMX result is the same in the even and the odd images (Z 24.8 and 23.6); an isolation cut on WADERs clusters makes WADERs worse (Z 27.1 -> 6.2), so the gain is from charge sharing, not from the cut. Caveats: dev/ndf 1.98 vs 1.74 (structure at 25-28 keV); sigma_xy of the new methods is a different width estimate from panaSKImg's fitted one and the 0.7 cut was not re-optimised.

Am-241 10-30 keV: constrained fit (Np L lines and intensity ratios from Lepy et al. 2008, Table 3) after the optimised cut - 29 images, gain correction per gain period



Am-241 10-30 keV: constrained fit (Np L lines and intensity ratios from Lepy et al. 2008, Table 3) - 29 images, gain correction per gain period, GMX (blend < 0.02) clustering



Injected objects in real Am-241 images (truth known)

Injected into real AMTEst images (3 images x 3 seeds): muons (15-60 px), electron worms, 59.5 keV diffuse blobs, 17.75 keV and Fe-55 X-rays, alphas, hot-column streaks, few-electron events, pairs of 5.9 keV X-rays 3 / 4 / 6 px apart. Success = one cluster with E within 5 % (pairs: both resolved). Same objects for every method.

EXT1 (0.12 e- read noise, low background)

method	muon	worm	blob	single	fe55	alpha	hotcol	lowE	pair3	pair4	pair6
WADERs	0.90	0.93	0.94	0.97	0.97	0.90	0.76	0.47	0.00	0.21	0.71
MFD+G	0.19	0.28	0.97	1.00	1.00	0.78	0.00	0.47	0.29	0.99	1.00
MHC	0.94	0.93	0.97	1.00	1.00	0.78	0.78	0.53	0.29	0.92	1.00
adaptive	0.94	0.93	0.97	1.00	1.00	0.89	0.78	0.53	0.29	0.92	1.00
GMX	0.93	0.93	0.96	0.99	0.97	0.94	0.74	0.64	0.75	0.93	0.93
adaptive+GMX	0.93	0.93	0.96	0.99	0.97	0.94	0.74	0.64	0.75	0.93	0.93

EXT3 (denser CCD)

method	muon	worm	blob	single	fe55	alpha	hotcol	lowE	pair3	pair4	pair6
WADERs	0.17	0.81	0.84	0.98	0.97	0.28	0.32	0.17	0.00	0.21	0.90
MFD+G	0.02	0.24	0.92	1.00	0.98	0.10	0.02	0.34	0.22	0.98	1.00
MHC	0.20	0.79	0.92	1.00	0.98	0.10	0.28	0.28	0.22	0.92	1.00
adaptive	0.19	0.79	0.90	1.00	0.98	0.26	0.30	0.28	0.22	0.92	1.00
GMX	0.20	0.81	0.89	0.97	0.97	0.28	0.25	0.40	0.78	0.98	1.00
adaptive+GMX	0.20	0.81	0.89	0.97	0.97	0.28	0.25	0.40	0.78	0.98	1.00

Few-electron events (1-20 e- in 1-5 pixels)

Simulated blank images (Poisson dark charge + 0.15 e- read noise, 2 x 1.7 Mpix per dark charge), 800 events of 1-20 e- spread over 1-5 pixels; efficiency by event charge, the fraction with exactly the right charge at 3 e-, and fake clusters per Mpix on a second blank with nothing injected.

dark charge 0.001 e-/pix, read noise 0.15 e- (simulated blank, 2 x 1.7 Mpix)

method	2 e-	3 e-	4 e-	5 e-	7 e-	10 e-	exact@3	fakes 2-5e/Mpix	1e/Mpix
WADERs	0.00	0.00	0.00	0.11	0.29	0.62	-	0.0	0
MFD+G 5sig	0.00	0.11	0.48	0.85	1.00	1.00	0.32	0.0	0
MFD+G 3sig count	0.22	0.84	0.97	1.00	1.00	1.00	0.91	0.3	4
MHC	0.01	0.81	0.97	0.99	1.00	1.00	0.97	0.0	0
adaptive	0.01	0.81	0.97	0.99	1.00	1.00	0.97	0.0	0
GMX	0.01	0.81	0.98	1.00	1.00	1.00	0.97	0.0	0
adaptive+GMX	0.01	0.81	0.98	1.00	1.00	1.00	0.97	0.0	0
DM-style >=1e-	1.00	1.00	1.00	1.00	1.00	1.00	0.91	7.3	1427

dark charge 0.01 e-/pix, read noise 0.15 e- (simulated blank, 2 x 1.7 Mpix)

method	2 e-	3 e-	4 e-	5 e-	7 e-	10 e-	exact@3	fakes 2-5e/Mpix	1e/Mpix
WADERs	0.00	0.00	0.00	0.11	0.30	0.56	-	0.0	0
MFD+G 5sig	0.00	0.00	0.02	0.18	0.95	1.00	-	0.0	0
MFD+G 3sig count	0.04	0.20	0.81	0.95	1.00	1.00	0.39	3.8	0
MHC	0.00	0.01	0.38	0.93	1.00	1.00	0.00	0.0	0
adaptive	0.00	0.01	0.38	0.93	1.00	1.00	0.00	0.0	0
GMX	0.00	0.01	0.38	0.93	1.00	1.00	0.00	0.0	0
adaptive+GMX	0.00	0.01	0.38	0.93	1.00	1.00	0.00	0.0	0
DM-style >=1e-	1.00	1.00	1.00	1.00	1.00	1.00	0.81	462.3	9425

dark charge 0.05 e-/pix, read noise 0.15 e- (simulated blank, 2 x 1.7 Mpix)

method	2 e-	3 e-	4 e-	5 e-	7 e-	10 e-	exact@3	fakes 2-5e/Mpix	1e/Mpix
WADERs	0.00	0.00	0.00	0.11	0.35	0.58	-	0.0	0
MFD+G 5sig	0.00	0.00	0.00	0.00	0.12	0.90	-	0.0	0
MFD+G 3sig count	0.02	0.06	0.28	0.67	0.98	1.00	0.18	36.3	1
MHC	0.00	0.00	0.00	0.04	0.56	0.99	-	0.0	0
adaptive	0.00	0.00	0.00	0.04	0.56	0.99	-	0.0	0
GMX	0.00	0.00	0.00	0.04	0.56	1.00	-	0.0	0
adaptive+GMX	0.00	0.00	0.00	0.04	0.56	1.00	-	0.0	0
DM-style >=1e-	1.00	1.00	1.00	1.00	1.00	1.00	0.58	7794.0	32173

Fe-55: injection and the real Ka peak (DM09 / DM10, EXT1 / EXT4, images

DM09 / DM10 (2025-08-27), EXT1 and EXT4, images 0-1: 220 injected 1580-e- clusters per width (median E_rec / E_true and the worst CCD), pairs of sigma-1.4-px clusters 5 px apart (fraction resolved with both within 5 %), and the real front-side K-alpha (mean / 5.899 keV and its spread over the 8 CCDs).

method	s0.3 med/worst	s0.8	s1.4	s1.8	pair5px	Ka/5.899	spread%
WADERs	0.9974/0.990	0.9843/0.943	0.9521/0.837	0.9214/0.737	0.00	0.9928	1.32
MFD+G	1.0001/1.000	1.0002/0.999	1.0001/0.996	0.9999/0.995	0.78	0.9968	0.29
MHC	1.0001/1.000	1.0002/0.999	1.0001/0.996	0.9998/0.995	0.77	0.9969	0.26
adaptive	1.0001/1.000	1.0002/0.999	1.0001/0.996	0.9997/0.995	0.77	0.9968	0.28
GMX	1.0000/1.001	0.9999/0.998	0.9996/0.994	0.9997/0.994	0.93	0.9968	0.23
adaptive+GMX	1.0000/1.001	0.9999/0.998	0.9996/0.994	0.9997/0.994	0.93	0.9968	0.23

Recommendations, corrections, files

Does A_GMX supersede the others?

Yes as the general clusterer. MHC is still slightly better on Fe-55 success in the Fe-55 data sets (0.964 vs 0.951) and on hot columns (0.78 vs 0.74), but A_GMX is better on crowded, noisy and high-dark-current images, worms, alphas, close and diffuse pairs, resolution, and 2-300x faster. Hybrids and the MHC-engine adaptive are superseded. MFD+G is kept for Fe-55 calibration, WADERs / B auto for the pipeline. All code and results are kept (see Files).

Corrections made during the study

AMTEst EXT4 was first called unusable, then baseline-shifted: both wrong - it is a genuinely crowded CCD (the raw .fz data show the same). The EXT3 'dark charge 0.5-13 e-/pix' of the early reports was the background map fitted on a crowded CCD. Real EXT1 'big events' are mostly horizontal 1-5-row streaks (median 1.4 keV). The adaptive (MHC engine) first used the image median dark charge for its occupancy (rows with high dark charge were called crowded; energies +1.4 % on DM10 img0) and the background offset in its noisy branch (-2.4 e- on PD06): both fixed.

Open items

inspect A_GMX labels and track merging by eye on real Am-241 tracks; exact pixel integrals for very sharp clusters; a background model for the 'unmodelled' PD06 images; re-optimise the sigma_xy cut for A_GMX on the crowded CCD and understand the 25-28 keV structure; decide whether A_GMX becomes an optional run_WADERs flag (needs approval).

Files

code: comm_scripts/clustering_tests/ (README); A_GMX document: docs/aGMX/aGMX_clustering.tex (Overleaf: docs/aGMX/aGMX_clustering_overleaf.zip); context: docs/CONTEXT_2026-10-24.md; results: ~/Documents/comm_scripts_data/session_2026-10-23_clustering/clustering_2026-10-25/.